

3-Hourly Solar Radiation for TCo1279-DART: Method & Status

Short briefing

AWI Dark-Doldrums Project

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The problem

The TCo1279-DART model runs (1950C and 2080C) save top-of-atmosphere shortwave radiation and cloud fractions every 3 hours, but save surface downward shortwave (**ssrd**) — the quantity the dark-doldrums study actually needs — only as a **monthly mean**. A monthly mean cannot show the short-lived low-wind/low-sun events the study is looking for. Running a real radiative transfer model to fill the gap isn't possible: it needs aerosol and humidity profiles that DART's saved output doesn't contain.

Approach: statistically reconstruct the missing 3-hourly **ssrd** from what DART *does* save, using ERA5 (which has both) to learn the relationship, then apply it to DART.

The method, briefly

Instead of predicting **ssrd** directly, we predict the normalized **clearness index** $k_t = ssrd/tisr$, where **tisr** is top-of-atmosphere incoming radiation — pure astronomy, computable exactly from lat/lon/time. Normalizing this way keeps the target in a stable [0,1]-ish range regardless of season, which is what lets a model trained on a few ERA5 months generalize at all.

Predictors: TOA transmission ($tsr/tisr$), solar zenith angle, and the four cloud fraction layers. We tested a plain cloud-fraction formula first and it only explained 57% of the variance; adding transmission and switching to gradient boosting (which lets clouds attenuate differently depending on cloud type, rather than one fixed sensitivity) raised that to **98.6%**, tested honestly on ERA5 months the model never saw during training (leave-one-month-out validation, RMSE 32.7 W/m², bias approx. 0).

After the 3-hourly prediction, each grid cell is **rescaled so its monthly mean exactly matches DART's own real monthly ssrd** — this removes any systematic level bias and guarantees consistency with DART's own numbers.

Applying it to DART: first test (January 1950)

Two things had to be solved specifically for DART, not needed for ERA5:

- **tisr lookup, not recomputation.** Since solar geometry only depends on day-of-year/time-of-day (not the specific year), we precomputed and independently verified two reusable reference files — one normal year, one leap year — covering every year in both DART periods exactly.

- **Correct accumulation window.** DART’s fields are 3-hour accumulations, confirmed three independent ways (metadata, and matching known $\sim 190 \text{ W/m}^2$ January climatology). Getting this wrong wouldn’t crash anything — it would silently distort the transmission predictor.

Result of the one-month test: before correction, the reconstruction was 6.6% too high (204 vs. 191 W/m^2 real DART monthly mean) — comparable to error levels seen during ERA5 validation. After the per-cell monthly rescale, the corrected output matches DART’s real monthly value almost exactly (residual 0.0002 W/m^2).

The open question

All the 98.6%-accuracy numbers above are *ERA5 predicting ERA5* — they prove the method captures how ERA5’s own radiation scheme behaves. They do **not** prove DART’s radiation scheme responds to the same cloud fraction in exactly the same way, since DART runs a different model at different resolution. The first hint this gap is real: per-cell correction factors on the test month ranged from $0.56\times$ to $5.3\times$, wider than seen in ERA5 (though the median is exactly 1.0 and 90% of cells fall in a much tighter 0.94–1.36 band — the extremes are a minority of outlier locations, not a global problem).

This can’t be closed directly — DART has no independent 3-hourly `ssrd` anywhere to check against; that’s the whole reason this method exists. The practical mitigation: the monthly rescale absorbs any *systematic* bias, so overall magnitude is trustworthy even where within-month shape might be imperfect.

Verdict and next steps

Not clear yet whether the model needs changing — one test month passing within the expected error range isn’t a failure signal, but the wide scale-factor spread needs to be checked, not ignored. Before committing to the full 240-month (1950C) + 156-month (2080C) run:

1. Map scale factors across a handful of contrasting months (not just one) and check whether large corrections cluster geographically (e.g. terrain, high-latitude winter) — a pattern would point to a fixable feature gap (e.g. adding surface albedo, which DART does carry monthly) rather than a fundamental problem.
2. Refactor for efficiency: the current pipeline re-reads the same `tisr` reference file every month even within the same calendar year. Restructuring to read it once per year cuts total reference-file reads from 396 to 33 before the full run is launched.
3. Once both checks pass, run the full two periods, one SLURM job per year.

Full technical report with all validation tables and figures available on request.